

Notes

- a. The flows for a recirculating unit assume the standard method of operation is recirculation throughout the treatment. It is recommended that the pipeline should be designed for open circuit operation to ensure adequate flow under all conditions.
- b. Clinical practice may require the inclusion of air during the treatment; it may also be necessary to switch to air in the unlikely event of an oxygen convulsion. Therefore consideration should be given to the provision of medical air from a separate dedicated medical air plant in accordance with [Chapter 7](#).
- c. Some hyperbaric chambers use air as a buffer and consequently less oxygen is consumed. The advice of the manufacturer should be sought. Where this is the case, the air should be supplied from a separate supply system complying with the requirements for medical air systems.

Nitrous oxide

- 4.28 Nitrous oxide is provided for anaesthetic purposes and occasionally for analgesic purposes. In all cases, each terminal unit should be capable of passing 15 L/min, but in practice the flow is unlikely to exceed 6 L/min.
- 4.29 When calculating diversities in a department, 15 L/min is allowed for the first and 6 L/min for the remainder, subject to the appropriate diversity factor being applied (see Table 15).
- 4.30 It is assumed that, for an operating department, nitrous oxide may be in use simultaneously in all

operating rooms. As it is unlikely that a patient would be anaesthetised in the anaesthetic room at the same time that a patient in the associated operating room was continuing to be treated under an anaesthetic (and because the duration of induction is short), no additional flow is included.

Nitrous oxide/oxygen mixture

- 4.31 All terminal units should be capable of passing 275 L/min for a very short period (normally of five seconds' duration) to supply inhalationary "gasps" by the patient, and a continuous flow of 20 L/min. The actual flow would not normally exceed 20 L/min.
- 4.32 The diversified flow in delivery rooms is based on 275 L/min for the first bed space and 6 L/min for each of the remainder, of which only half of the women in labour will be using gas for 50% of the time. (The peak inhalationary "gasp" is 275 L/min, whereas the respirable minute volume will be catered for with a flow of 6 L/min – it should also be borne in mind that a woman in labour would not continuously breathe the analgesic mixture.) For larger maternity departments with twelve or more LDRP rooms, two peak inhalationary "gasps" are included.
- 4.33 Nitrous oxide/oxygen mixture may be used in other areas for analgesic purposes. The diversified flow is based on 10 L/min for the first treatment space, and 6 L/min for a quarter of the remainder for 25% of the time.
- 4.34 Design and diversified flows for nitrous oxide/oxygen mixtures are given in [Table 16](#).

Table 15 Nitrous oxide: design and diversified flows

Department	Design flow for each terminal unit (L/min)	Diversified flow Q (L/min)
Accident & emergency: resuscitation room, per trolley space	10	$Q = 10 + [(n - 1)6/4]$
Operating	15	$Q = 15 + (nT - 1)6$
Maternity: operating suites	15	$Q = 15 + (nS - 1)6$
Radiological: all anaesthetic and procedures rooms	15	$Q = 10 + [(n - 1)6/4]$
Critical care areas	15	$Q = 10 + [(n - 1)6/4]$
Oral surgery/orthodontic: consulting rooms, type 1	10	$Q = 10 + [(n - 1)6/4]$
Other departments	10	No additional flow included
Equipment service rooms	15	No additional flow included